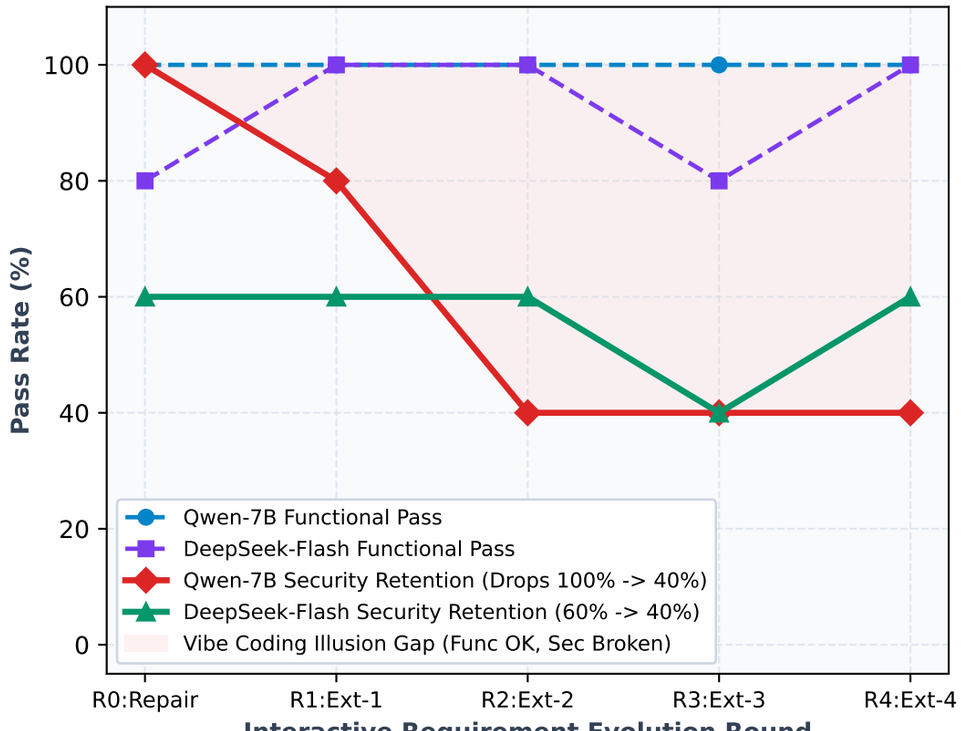
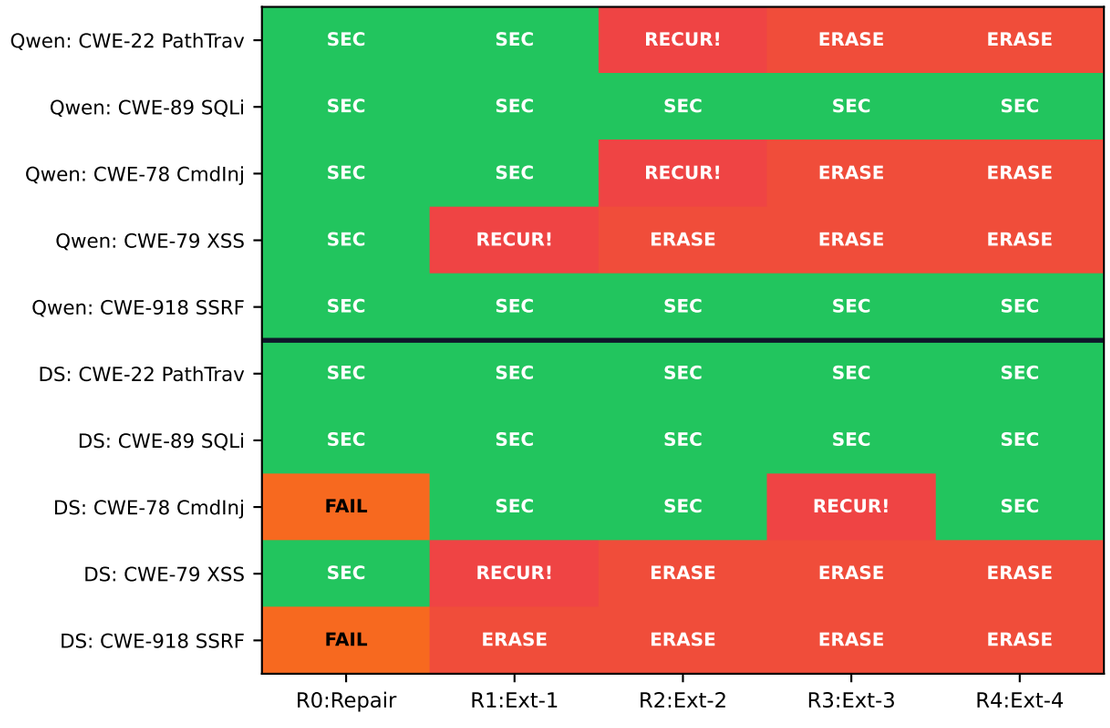
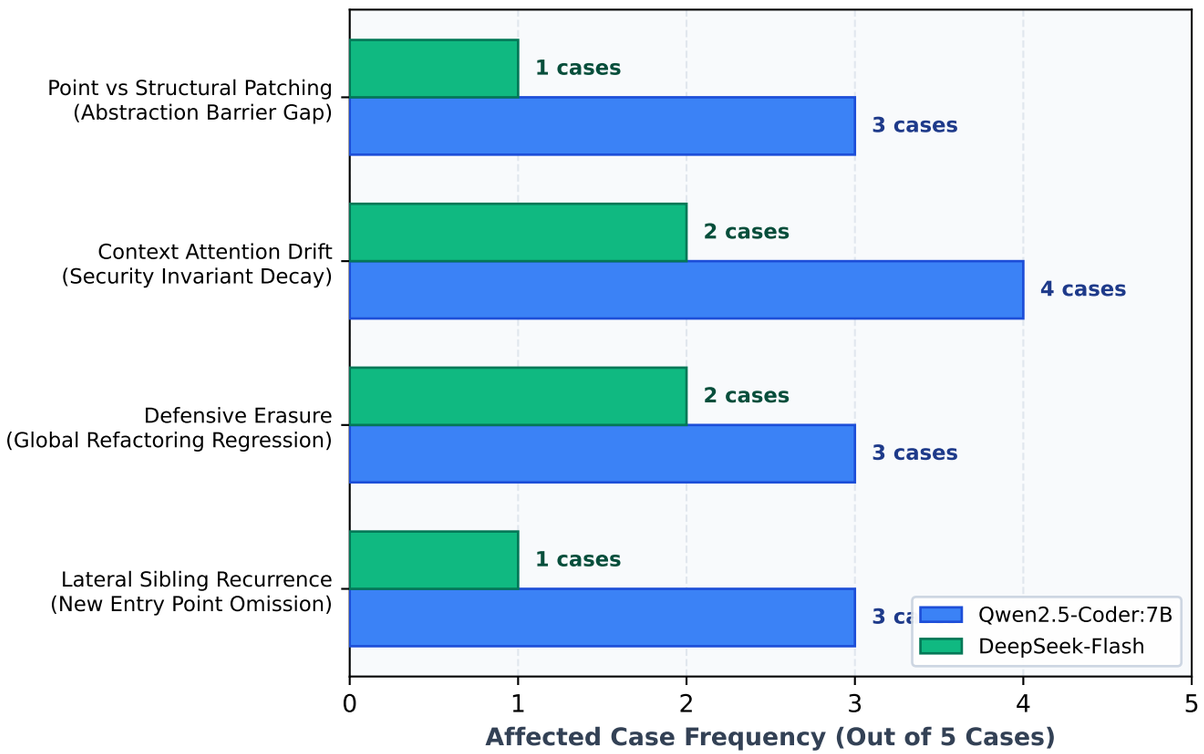
A. Longitudinal Decoupling: Functionality vs Security Retention



B. Multi-Turn Vulnerability State Matrix (Qwen-7B vs DeepSeek)



C. Recurrence Phenomenon Taxonomy & Failure Modes



D. Mechanism Dynamics & PACG Architectural Solution

Observed Vulnerability Recurrence Lifecycle

[R0] Prompt: 'Fix CWE-22' -> LLM patches target file -> Security probe PASS

[R1~R2] Prompt: 'Add delete_file' -> LLM creates sibling func without guard -> RECUR!

[R3~R4] Prompt: 'Add preview' -> LLM refactors entire file -> Erases earlier guards -> FAIL

Proposed Defense: AST Invariant Pinning & PACG Gateway

1. Invariant Pinning: Extract AST security guards (e.g. resolve_path) as persistent invariants
2. Sibling Cross-Check: Intercept new entry points & enforce inherited security contracts
3. PACG Runtime Gateway: Execute dynamic security probes before commit to developer
4. Verification Output: Closed-loop repair without developer cognitive burden